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### Baluster connectors and systems having the same

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#### Abstract

Baluster connectors and railing systems utilizing baluster connectors. The baluster connectors can include a baluster engaging portion, a support engaging portion, and a flange.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. provisional application No. 63/145,574, filed on Feb. 4, 2021, entitled: “BALUSTER CONNECTORS AND SYSTEMS HAVING THE SAME.” The entirety of the aforementioned application is incorporated by reference herein.

### TECHNICAL FIELD

(1) The present disclosure relates to baluster connectors and/or to systems that utilize the baluster connectors.

### BACKGROUND

(2) Certain conventional railing systems utilize baluster coupling components that require the use of a separate fastener to secure the coupling component to a rail and/or to a baluster. However, such systems can require additional or specialized tools to install, and can require additional installation steps that may result in installation errors and/or a longer or more complicated installation process.

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Illustrative aspects are described in detail below with reference to the attached drawing figures, which are incorporated by reference herein and wherein:

- (2) FIG. 1 depicts an example railing system, in accordance with aspects hereof;
- (3) FIGS. 2A-2C depict an example baluster connector, in accordance with aspects hereof;
- (4) FIG. 2D depicts another example baluster connector, in accordance with aspects hereof;
- (5) FIGS. 3A-3C depict another example baluster connector, in accordance with aspects hereof;
- (6) FIGS. 4A-4C depict yet another example baluster connector, in accordance with aspects hereof;
- (7) FIG. 4D depicts another example railing system, in accordance with aspects hereof;
- (8) FIGS. 5A-5C depict another example baluster connector, in accordance with aspects hereof;
- (9) FIGS. 6A-6C depict yet another example baluster connector, in accordance with aspects hereof;
- (10) FIGS. 7A-7C depict another example baluster connector, in accordance with aspects hereof;
- (11) FIGS. 8A-8C depict yet another example baluster connector, in accordance with aspects hereof;
- (12) FIGS. 9A-9C depict another example baluster connector, in accordance with aspects hereof;

(13) FIGS. **10A-10C** depict yet another example baluster connector, in accordance with aspects hereof; and

(14) FIG. **11** depicts another example railing system, in accordance with aspects hereof.

#### DESCRIPTION

(15) Aspects herein relate to baluster connectors. Additional aspects herein relate to systems that utilize baluster connectors for coupling a baluster to one or more supports, e.g., a rail. As discussed above, certain conventional baluster coupling components can require the use of a separate fastener to secure the coupling component to a rail and/or to a baluster. However, such conventional systems can require additional or specialized tools to install, and can require additional installation steps. There is a need for baluster connectors that do not require additional fasteners or other components to couple a baluster to a rail, and/or that offer ease of installation.

(16) The baluster connectors and railing systems disclosed herein can alleviate one or more of the problems described above. For instance, in aspects, the baluster connectors disclosed herein can include a baluster engaging portion that can be coupled to a baluster, e.g., inserted into the baluster, and a support engaging portion that can be coupled to a support, e.g., inserted into a void in a support and/or a rail. In such aspects, the baluster connectors disclosed herein can be secured to a support and to the baluster without the use of additional separate fasteners or specialized tools, thereby allowing for ease of installation.

(17) Accordingly, in one aspect a baluster connector is provided. The baluster connector can include a first end and a second end opposite the first end. The baluster connector can also include a flange positioned between the first end and the second end. The baluster connector can further include a baluster engaging portion, where the baluster engaging portion extends out from the flange to the first end. The baluster connector can also include a support engaging portion, where the support engaging portion extends out from the flange to the second end.

(18) In yet another aspect, a railing system is provided. The railing system can include a top rail, the top rail extending from a first end to a second end; and a bottom rail, the bottom rail extending from a first end to a second end, where the bottom rail is vertically spaced apart from the top rail. The railing system can also include one or more balusters, each of the one or more balusters including a first end and a second end. The railing system can also include one or more baluster connectors. The one or more baluster connectors are adapted for coupling to: i) the second end of one of the one or more balusters; and ii) the bottom rail. Each of the one or more balusters includes a first end and a second end opposite the first end. Each of the one or more baluster connectors can also include a flange positioned between the first end and the second end. Each of the one or more baluster connectors can further include a baluster engaging portion, where the baluster engaging portion extends out from the flange to the first end. Each of the one or more baluster connectors can also include a support engaging portion, where the support engaging portion extends out from the flange to the second end.

(19) As used herein, the term “about” means plus or minus 10% of the numerical value of the number with which it is being used unless otherwise indicated or custom in the art dictates otherwise. Therefore, for example, “about 10 millimeters (mm)” includes the value 10 mm in addition to the range of 9 mm to 11 mm.

(20) Turning to the figures, and FIG. **1** in particular, an example railing system **100** is depicted. While the railing system **100** is depicted as a level railing system, it should be understood that other types of railing systems are also contemplated for use with the systems disclosed herein, including but not limited railing systems for installation on an angled surface, e.g., slopes and/or stairways.

(21) The railing system **100** depicted in FIG. **1** includes a top rail **110** and a bottom rail **120**, a plurality of balusters **130**, and a plurality of baluster connectors **200**. As can be seen in the aspect depicted in FIG. **1**, the top rail **110** and the bottom rail **120** can be vertically spaced apart, with the plurality of balusters **130** extending between the top rail **110** and the bottom rail **120**. As will be discussed below in more detail, each of the plurality of baluster connectors **200** can couple one end

of a baluster to the bottom rail **120** or other support. In an aspect not depicted in FIG. **1**, the baluster connectors disclosed herein can be utilized to couple a baluster to a top rail, e.g., the top rail **110**, in addition to a baluster connector being utilized to couple a baluster to a bottom rail, e.g., the bottom rail **120**, or instead of utilizing a baluster connector to couple a baluster to a bottom rail.

(22) It should be understood that, as used herein, the terms top rail and bottom rail may not exclusively refer to a top-most rail and a bottom-most rail, respectively, of a railing system. Rather, in certain aspects, a baluster may extend between a top rail and a bottom rail, where such a top rail and/or such a bottom rail is an intermediate rail in a railing system. For instance, in one specific example, a railing system may include a bottom rail (or other bottom support), a top rail, and an intermediate rail positioned between the top and bottom rails. In such an example, a baluster may be coupled to the bottom rail (or other bottom support) and the intermediate rail. In this example, the intermediate rail may also be referred to as a top rail. In another example, a baluster may be coupled to a top rail and to an intermediate rail, with a bottom rail positioned beneath the intermediate rail. In such an example, the intermediate rail may be referred to as a bottom rail. In yet another example, a baluster may be coupled to two intermediate rails with top and bottom rails positioned on either side of the two intermediate rails. In such an example, the two intermediate rails coupled to the baluster may be referred to as top and bottom rails. Accordingly, while in aspects, the terms top rail and bottom rail may be utilized, it should be understood that the terms top rail and bottom rail may refer to the rails with which the opposing ends of the baluster may be coupled to, which may not necessarily be the top-most and bottom-most rails of a railing system. Further, as discussed above, the disclosure herein also contemplates that a baluster connector can be utilized to connect a baluster to a support other than a bottom rail. In one or more aspects, the support structure can be a floor, ground, step, ramp, or the like, and in such an aspect, a baluster connector, e.g., via a support engaging portion, can interact with a void or aperture in such a support structure in a manner similar to that described herein with respect to a bottom rail.

(23) In aspects, any or all of the components of the railing system **100** can be comprised of metal, including but not limited to iron, steel, and/or aluminum; polymeric material; wood; filler material; or a combination thereof.

(24) In the aspect depicted in FIG. **1**, the top rail **110** extends between a first end **112** and a second end **114** in a direction that is the same or similar to the direction of extension between a first end **122** and a second end **124** of the bottom rail **120**, when the railing system is in an assembled configuration. In the same or alternative aspects, when assembled, the top rail **110** and the bottom rail **120** can be parallel or substantially parallel with one another. As used herein, a railing system in an assembled configuration can refer to a baluster being coupled to a top rail, e.g., the top rail **110**, and to the bottom rail, e.g., the bottom rail **120**, as depicted in FIG. **1** for the baluster **136**, including the use of a baluster connector, e.g., the baluster connector **210a**, to couple the baluster **136** to the bottom rail **120**.

(25) In various aspects, such as that depicted in FIG. **1**, the top rail **110** and/or the bottom rail **120** can include a void **113** or **121**, respectively, extending through at least a portion of the top rail **110** and/or the bottom rail **120**. In one aspect, the void **113** and/or the void **121** can extend through the length  $l_{sub.1}$  of the top rail **110** and/or the length  $l_{sub.2}$  of the bottom rail **120**, respectively. In aspects, the voids **113** and/or **121** can provide space for the insertion of the respective portions of the baluster and baluster connectors, respectively, as discussed below.

(26) In an alternative aspect, the top rail **110** and/or the bottom rail **120** can be solid or partly solid components, e.g., do not include a void **113** and/or a void **121** that extends through the length  $l_{sub.1}$  of the top rail **110** and/or the length  $l_{sub.2}$  of the bottom rail **120**, respectively. In such an aspect, the top rail **110** and/or bottom rail **120** can include discrete voids for receiving respective portions of the baluster and/or baluster connectors, such as the aperture **111** and associated void shown in phantom in the top rail **110** for receipt of a portion of the baluster **132**.

(27) As can be seen in FIG. **1**, the plurality of balusters **130** are depicted as having a cylindrical

shape and/or exhibit a circular cross-sectional shape in a plane transverse to their length,  $l$ . It should be understood that the plurality of balusters can be any shape, including a shape exhibiting a square or rectangular cross-sectional shape. In aspects, the plurality of balusters **130** can exhibit a length  $l$  of between about 30 centimeters (cm) and about 150 cm. In the same or alternative aspects, the plurality of balusters can exhibit a width  $w$  of between about 2.5 millimeters (mm) to about 110 mm, from about 5 mm to about 80 mm, from about 6 mm to about 65 mm, or from about 7 mm to about 50 mm.

(28) In certain aspects, as can be seen in FIG. 1, a first end **133** of the baluster **132** can be inserted into an aperture **111** of the top rail **110** and allowing the first end **133** of the baluster **132** to extend into the void **113**. In the same or alternative aspects, the second end **135** of the baluster **132** can be coupled to the bottom rail **120** via the baluster connector **210**. In aspects, a baluster engaging portion **220** of the baluster connector **210** can be inserted into a void at or near the second end **135** of the baluster **132**, and the support engaging portion **230** of the baluster connector **210** can be inserted into an aperture **123** of the bottom rail **120**, thereby coupling the baluster **132** to the bottom rail **120**. As can be seen in FIG. 1, the baluster connector **210b** includes a baluster engaging portion **220b** inserted into the baluster **134** while the support engaging portion **230b** is inserted into the bottom rail **120**, e.g., extending into the void **121**. As discussed further below, with reference to FIG. 11, in certain aspects, the baluster connectors disclosed herein, may be utilized to secure a baluster to a top rail in addition to, or instead of, a bottom rail.

(29) FIGS. 2A-2C depict the baluster connector **210**. In aspects, as discussed above, the baluster connector **210** includes a baluster engaging portion **220** and a support engaging portion **230**. As best seen in the aspect depicted in FIG. 2A, the baluster connector **210** includes a flange **240** that is positioned between a first end **221** and a second end **223**. In aspects, the baluster engaging portion **220** can extend out from the flange **240** to the first end **221**. In the same or alternative aspects, the support engaging portion **230** can extend out from the flange **240** to the second end **223**. In aspects, the baluster connector **210** can be one integral component, e.g., where the baluster engaging portion **220**, the support engaging portion **230**, and the flange **240** are integral. In one or more aspects, the baluster connector **210** can be formed using any convenient manufacturing techniques, such as injection molding or extrusion molding.

(30) It should be understood that the specific shape of the baluster connector **210** is just one example shape and that other shapes are contemplated for use in the present disclosure. For instance, in one alternative aspect, the baluster engaging portion **220** can exhibit a cross-sectional shape of a square or rectangle to accommodate an appropriate fit with a similarly shaped baluster. In another alternative aspect, the baluster engaging portion **220** can be shaped as depicted in FIG. 2A and a separate adapter member can be coupled to the baluster engaging portion **220** to accommodate a non-cylindrical shaped baluster, e.g., a baluster having a square or rectangular cross-sectional shape.

(31) In various aspects, the first end **221** can be positioned a distance  $h_{sub.1}$  from the flange **240** by about 2 millimeters (mm) to about 50 mm, by about 3 mm to about 45 mm, or by about 4 mm to about 40 mm. In the same or alternative aspects, the second end **223** can be positioned a distance  $h_{sub.2}$  from the flange **240** by about 2 mm to about 50 mm, by about 3 mm to about 45 mm, or by about 4 mm to about 40 mm. In aspects, the length of the baluster engaging portion **220** can be the distance  $h_{sub.1}$  between the first end **221** and the flange **240**. In various aspects, the length of the support engaging portion **230** can be the distance  $h_{sub.2}$  between the second end **223** and the flange **240**.

(32) In certain aspects, the baluster engaging portion **220** can include a sidewall **222** that extends out from the flange **240** to the first end **221**, as best seen in FIGS. 2A and 2C. In various aspects, the baluster engaging portion **220** has a width  $w_{sub.1}$  that is from about 2.5 mm to about 110 mm, from about 5 mm to about 80 mm, from about 6 mm to about 65 mm, or from about 7 mm to about 50 mm. In the same or alternative aspects, the baluster engaging portion **220** exhibits a

substantially similar width  $w_{sub.1}$  from a position adjacent the flange **240** to a position at or near the first end **221**. In such an aspect, the sidewall **222** can define a cylinder-shaped structure having a substantially similar diameter from end to end. In an alternative aspect not depicted in the figures, the baluster engaging portion or associated sidewall can have a tapered (or other) shape to allow for ease of engagement, e.g., insertion into a void of a baluster. In the aspect depicted in FIGS. 2A-2C, the baluster engaging portion **220** and the support engaging portion **230** are depicted as having voids in the respective portions. In alternative aspects, the baluster engaging portion **220** and/or the support engaging portion **230** can lack such voids and can include one or more materials forming a solid structure for all or a portion of the width  $w_{sub.1}$  or the width  $w_{sub.2}$ , respectively.

(33) In aspects, as discussed above, the baluster engaging portion **220** can be inserted into a void of a baluster, e.g., the baluster **132** of FIG. 1. In aspects, when assembled, the baluster engaging portion **220** may be entirely or partially positioned within the baluster **132**. In aspects, such as that depicted in FIG. 1, where the first end **133** of the baluster **132** is inserted into the aperture **111**, the first end **133** may shift vertically within the void **113** or aperture **111** of the top rail **110**, since the first end **133** may not be fixedly secured to the top rail **110**. In such aspects, in order for the baluster **132** to not become disengaged or decoupled from the bottom rail **120** and/or from the baluster connector **210**, a distance  $h_{sub.1}$  of the baluster engaging portion **220** of the baluster connector **210** can be greater than or equal to the distance the first end **133** of the baluster **132** can vertically shift within the void **113** or the aperture **111**, when in an assembled configuration.

(34) In an alternative aspect not depicted in the figures, a baluster can be received in a void of the baluster engaging portion as opposed to the baluster engaging portion of a baluster connector being inserted into a void of the baluster.

(35) In aspects, the support engaging portion **230** can engage with or couple to the bottom rail **120**, or other support, in any convenient manner. In aspects, the support engaging portion **230** can include one or more structural features that can contact at least a portion of a bottom rail, e.g., the bottom rail **120** of FIG. 1, to temporarily or permanently secure the baluster connector **210** to the bottom rail **120** or other support. For instance, as best seen in FIG. 2A, the support engaging portion **230** includes a plurality of ridges **232** that can provide a friction fit and/or compression fit in or to the bottom rail **120** at or near an aperture of the bottom rail, e.g., the aperture **123** of the bottom rail **120**. In various aspects, all or a portion of the plurality of ridges **232** can interact with one or more portions of the bottom rail **120** to provide a friction fit and/or compression fit, e.g., the portions of the bottom rail **120** that define an aperture, e.g., the aperture **123**, in the bottom rail. In such aspects, all or a portion of the plurality of ridges **232** can contact one or more portions of the bottom rail **120** that define the aperture **123**. In certain aspects, the width  $w_{sub.2}$  of the support engaging portion **230** can be equal to or greater than a width and/or diameter of the aperture **123** or void so as to facilitate such a compression and/or friction fit. In various aspects, the support engaging portion **230** can have a width  $w_{sub.2}$  that is from about 2.5 mm to about 110 mm, from about 5 mm to about 80 mm, from about 6 mm to about 65 mm, or from about 7 mm to about 50 mm.

(36) As discussed above, in the aspect depicted in FIG. 2A, the baluster connector **210** can include a plurality of ridges **232** on the support engaging portion **230**, which may facilitate temporarily or permanently securing the baluster connector **210** to the bottom rail **120**. In the aspect depicted in FIG. 2A, the plurality of ridges **232** are circular ridges or serrations, e.g., the ridges or serrations are parallel or substantially parallel to one another. In aspects, such circular ridges can provide a tight fit with a support or bottom rail and/or lock in place the support engaging portion **230**, and/or the baluster connector **210**, to the bottom rail or support. In the same or alternative aspects, the circular ridges may function as a seal to restrict the ability of water or other elements to enter into the bottom rail or support. In an alternate aspect not depicted in the figures, the plurality of ridges or serrations may be helical, e.g., spiral, which may aid in inserting the support engaging portion **230** into the support or bottom rail, e.g., by rotating the plug into an aperture of the support

or bottom rail.

(37) In certain aspects, the plurality of ridges **232** may be integral to the support engaging portion **230**. In alternate aspects, the plurality of ridges **232** may be provided as an over mold, e.g., formed from rubber or other material. In one aspect, the plurality of ridges **232** may be an over mold material that includes two, three, four, five, six, seven, or eight ridges. FIG. 2D depicts an example variation of the support engaging portion **230** of the baluster connector **210**, designated as baluster connector **210a**. In the aspect depicted in FIG. 2D, support engaging portion **230a** of the baluster connector **210a** includes a plurality of ridges **232a** that are present on a portion of the length of the support engaging portion **230a**. In such aspects, not having ridges along the full length of the support engaging portion **230a** may result in increased production efficiencies and/or reduce the amount of material necessary to produce the baluster connectors **210a**.

(38) In aspects, when a baluster connector couples a baluster to the bottom rail, the flange may be positioned so that it is in contact with an outer surface of the bottom rail. For instance, as can be seen in FIG. 1, the flange **240a** of the baluster connector **210a** can be in contact with an outer surface **125** of the bottom rail **120** when the baluster connector **210a** is coupling and/or connecting the baluster **136** to the bottom rail **120**. It should be understood that, while the flange **240a** of the baluster connector **210a** depicted in FIG. 1, exhibits a flat-disc-shaped appearance, other example flanges are also contemplated for the baluster connectors disclosed herein. For instance, in an aspect not depicted in the figures, the flange, e.g., the flange **240a**, can exhibit an angled or tapered shape, where a portion of the tapered region can be inserted into the aperture **123** of the bottom rail **120** to facilitate a compression fit and/or to function as a seal to prevent debris or water ingress into the bottom rail **120**. It should also be understood that, when assembling any of the railing systems described herein with any of the baluster connectors described herein an O-ring, caulk, adhesive, or over molding applied to a portion of a baluster connector, or a combination thereof, can be utilized to provide better baluster connector retention, water ingress protection, and/or baluster “rattling” prevention.

(39) As best seen in the aspect depicted in FIGS. 2B and 2C, the baluster connector **210** can optionally include a through-hole **250** that extends from the baluster engaging portion **220** to the support engaging portion **230**, such that an interior portion of the baluster engaging portion **220** is in fluid communication with an interior portion of the support engaging portion **230**. In such an aspect, the through-hole **250** may reduce the mass of the baluster connector and/or allow moisture transmission there through. In an alternate aspect, the baluster connector may not include such a through-hole thereby closing off all or part of an interior portion of the baluster engaging portion **220** to an interior portion of the support engaging portion **230**. For instance, in one alternative aspect, the flange **240** may structurally resemble a flat, solid disc shaped component, and/or may not include a void, which in aspects, may limit moisture transport and/or pest or insects from traversing there through.

(40) FIGS. 3A-3C depict another baluster connector **300**. In aspects, the baluster connector **300** includes a baluster engaging portion **320**, a support engaging portion **330**, and a flange **340**. In aspects, the baluster connector **300** can operate similarly to the baluster connector **210** discussed above with reference to FIGS. 1 and 2A-2C. For instance, in aspects, the baluster engaging portion **320** can be inserted into a void of a baluster, e.g., a void in the second end **135** of the baluster **132** of FIG. 1, while the support engaging portion **330** can be coupled to a bottom rail, e.g., the bottom rail **120** of FIG. 1. In the same or alternative aspects, when in the assembled configuration, the flange **340** of the baluster connector **300** may contact an outer surface of the bottom rail, e.g., the outer surface **125** of the bottom rail **120** of FIG. 1.

(41) As best seen in the aspect depicted in FIG. 3A, the baluster **310** includes a flange **340** that is positioned between a first end **321** and a second end **323**. In aspects, the baluster engaging portion **320** can extend out from the flange **340** to the first end **321**. In the same or alternative aspects, the support engaging portion **330** can extend out from the flange **340** to the second end **323**.



(42) In various aspects, any or all of the dimensional properties of the baluster connector **210** of FIGS. 2A-2C discussed above can apply to the baluster connector **300**. For instance, the length of the baluster engaging portion **320** and/or the distance h.sub.1 from the first end **321** to the flange **340**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**. In the same or alternative aspects, the length of the support engaging portion **330** and/or the distance h.sub.2 from the second end **323** to the flange **340**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**.

(43) In certain aspects, the baluster engaging portion **320** can include a sidewall **322** that extends from the flange **340** to the first end **321**, as best seen in FIG. 3A. In various aspects, the baluster engaging portion **320** has a width w.sub.1 that can include any or all of the values and ranges of the width w.sub.1 of the baluster engaging portion **220** of the baluster connector **210** discussed above.

(44) In aspects, the support engaging portion **330** can include one or more structural features that can contact at least a portion of a bottom rail, e.g., the bottom rail **120** of FIG. 1 to temporarily or permanently secure the baluster connector **300** to a bottom rail. For instance, in the aspect seen in FIG. 3A, the support engaging portion **330** includes a plurality of angled ridges **332** that can provide a friction fit and/or compression fit in a bottom rail, e.g., the bottom rail **120** at or near the aperture **123** of FIG. 1. In such aspects, the width w.sub.2 of the support engaging portion **330** can be equal to or greater than a width of the aperture **123** or void so as to facilitate such a compression and/or friction fit. In various aspects, the support engaging portion **330** can have a width w.sub.2 that includes any of the values and/or ranges of the support engaging portion **230** of the baluster connector **210** of FIG. 2A discussed above.

(45) FIGS. 4A-4C depict another baluster connector **400**. In aspects, the baluster connector **400** includes a baluster engaging portion **420**, a support engaging portion **430**, and a flange **440**. In aspects, the baluster connector **400** can operate similarly to the baluster connector **210** discussed above with reference to FIGS. 1 and 2A-2C. For instance, in aspects, the baluster engaging portion **420** can be inserted into a void of a baluster, e.g., the void in the second end **135** of the baluster **132** of FIG. 1, while the support engaging portion **430** can be coupled to a bottom rail, e.g., the bottom rail **120** of FIG. 1. In the same or alternative aspects, when in the assembled configuration, the flange **440** of the baluster connector **400** may contact an outer surface of the bottom rail, e.g., the outer surface **125** of the bottom rail **120** of FIG. 1.

(46) As best seen in the aspect depicted in FIG. 4A, the baluster **410** includes a flange **440** that is positioned between a first end **421** and a second end **423**. In aspects, the baluster engaging portion **420** can extend out from the flange **440** to the first end **421**. In the same or alternative aspects, the support engaging portion **430** can extend out from the flange **440** to the second end **423**.

(47) In various aspects, one or more of the dimensional properties of the baluster connector **210** of FIGS. 2A-2C discussed above can apply to the baluster connector **400**. For instance, the length of the baluster engaging portion **420** and/or the distance h.sub.1 from the first end **421** to the flange **440**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**. In the same or alternative aspects, the length of the support engaging portion **430** and/or the distance h.sub.2 from the second end **423** to the flange **440**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**.

(48) In certain aspects, the baluster engaging portion **420** can include a sidewall **422** that extends from the flange **440** to the first end **421**, as best seen in FIG. 4A. In various aspects, the baluster engaging portion **420** has a width w.sub.1 that can include any or all of the values and ranges of the width w.sub.1 of the baluster engaging portion **220** of the baluster connector **210** discussed above. In various aspects, the support engaging portion **430** can have a width w.sub.2 that includes any of the values and/or ranges of the support engaging portion **230** of the baluster connector **210** of FIG. 2A discussed above.

(49) In aspects, the support engaging portion **430** can include one or more structural features that can contact at least a portion of a bottom rail, e.g., the bottom rail **120** of FIG. **1** to temporarily or permanently secure the baluster connector **400** to a bottom rail. For instance, as best seen in FIG. **4A**, the support engaging portion **430** includes a plurality of spaced apart locking members **432** that can interact with a portion of a bottom rail, a complementary locking member associated with a bottom rail, or both. For instance, in an aspect of a railing system **460** depicted in FIG. **4D**, the support engaging portion **430** can be inserted into an aperture of a bottom rail, e.g., the aperture **463** of the bottom rail **462**, and a portion of one or more of the plurality of spaced apart locking members **432**, e.g., the portion **433** depicted in FIG. **4A**, can interact with an inner surface **461** of the bottom rail **462** (and/or a complementary locking member associated with the bottom rail **462** not depicted in FIG. **4D**) in order to secure the baluster connector **400** in place and to the bottom rail **462**. In aspects, for example, the portion **433** of one of the plurality of spaced apart locking members **432** may interact with the bottom rail **462**, e.g., via the inner surface **461**, in such a manner so as to physically prevent disengagement or decoupling of the baluster connector **400** (or baluster coupled thereto) from the bottom rail **462** when an upward force is applied to the baluster connector **400** or baluster coupled thereto.

(50) As best seen in the aspect depicted in FIGS. **4B** and **4C**, the baluster connector **400** can optionally include a through-hole **450** that extends from the baluster engaging portion **420** to the support engaging portion **430**, such that an interior portion of the baluster engaging portion **420** is in fluid communication with an interior portion of the support engaging portion **430**. In an alternate aspect, the baluster connector may not include such a through-hole thereby closing off all or part of an interior portion of the baluster engaging portion **420** to an interior portion of the support engaging portion **430**. For instance, in one alternative aspect, the flange **440** may structurally resemble a flat, solid disc shaped component, and/or may not include a void.

(51) FIGS. **5A-5C** depict another baluster connector **500**. In aspects, the baluster connector **500** includes a baluster engaging portion **520**, a support engaging portion **530**, and a flange **540**. In aspects, the baluster connector **500** can operate similarly to the baluster connector **210** discussed above with reference to FIGS. **1** and **2A-2C**. For instance, in aspects, the baluster engaging portion **520** can be inserted into a void of a baluster, e.g., the void at or near a second end **135** of the baluster **132** of FIG. **1**, while the support engaging portion **530** can be coupled to a bottom rail, e.g., the bottom rail **120** of FIG. **1**. In the same or alternative aspects, when in the assembled configuration, the flange **540** of the baluster connector **500** may contact an outer surface of the bottom rail, e.g., the outer surface **125** of the bottom rail **120** of FIG. **1**.

(52) As best seen in the aspect depicted in FIG. **5A**, the baluster **510** includes a flange **540** that is positioned between a first end **521** and a second end **523**. In aspects, the baluster engaging portion **520** can extend out from the flange **540** to the first end **521**. In the same or alternative aspects, the support engaging portion **530** can extend out from the flange **540** to the second end **523**.

(53) In various aspects, one or more of the dimensional properties of the baluster connector **210** of FIGS. **2A-2C** discussed above can apply to the baluster connector **500**. For instance, the height of the baluster engaging portion **520** and/or the distance  $h_{sub.1}$  from the first end **521** to the flange **540**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**. In the same or alternative aspects, the height of the support engaging portion **530** and/or the distance  $h_{sub.2}$  from the second end **523** to the flange **540**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**.

(54) In certain aspects, the baluster engaging portion **520** can include a sidewall **522** that extends from the flange **540** to the first end **521**, as best seen in FIG. **5A**. In various aspects, the baluster engaging portion **520** has a width  $w_{sub.1}$  that can include any or all of the values and ranges of the width  $w_{sub.1}$  of the baluster engaging portion **220** of the baluster connector **210** discussed above. In various aspects, the support engaging portion **530** can have a width  $w_{sub.2}$  that includes any of

the values and/or ranges of the support engaging portion **230** of the baluster connector **210** of FIG. 2A discussed above.

(55) In aspects, the support engaging portion **530** can include one or more structural features that can contact at least a portion of a bottom rail, e.g., the bottom rail **120** of FIG. 1 to temporarily or permanently secure the baluster connector **500** to a bottom rail. For instance, as best seen in the aspect depicted in FIG. 5A, the support engaging portion **530** exhibits a variable width shape having a maximum width portion **533** positioned about midway along the length of the support engaging portion **530**. In such aspects, this maximum width portion **533** can interact with a portion of a bottom rail, a complementary locking member associated with a bottom rail, or both, in a manner similar to that described above in FIG. 4D with respect to the baluster connector **400**. In aspects, for example, the maximum width portion **533** may interact with the bottom rail **462**, e.g., via the inner surface **461**, in such a manner so as to physically prevent disengagement or decoupling of the baluster connector **500** (or baluster coupled thereto) from the bottom rail **462** when an upward force is applied to the baluster connector **500** or baluster coupled thereto.

(56) FIGS. 6A-6C depict another baluster connector **600**. In aspects, the baluster connector **600** includes a baluster engaging portion **620**, a support engaging portion **630**, and a flange **640**. In aspects, the baluster connector **600** can operate similarly to the baluster connector **210** discussed above with reference to FIGS. 1 and 2A-2C. For instance, in aspects, the baluster engaging portion **620** can be inserted into a void of a baluster, e.g., the void at or near the second end **135** of the baluster **132** of FIG. 1, while the support engaging portion **630** can be coupled to a bottom rail, e.g., the bottom rail **120** of FIG. 1. In the same or alternative aspects, when in the assembled configuration, the flange **640** of the baluster connector **600** may contact an outer surface of the bottom rail, e.g., the outer surface **125** of the bottom rail **120** of FIG. 1.

(57) As best seen in the aspect depicted in FIG. 6A, the baluster **610** includes a flange **640** that is positioned between a first end **621** and a second end **623**. In aspects, the baluster engaging portion **620** can extend out from the flange **640** to the first end **621**. In the same or alternative aspects, the support engaging portion **630** can extend out from the flange **640** to the second end **623**.

(58) In various aspects, one or more of the dimensional properties of the baluster connector **210** of FIGS. 2A-2C discussed above can apply to the baluster connector **600**. For instance, the height of the baluster engaging portion **620** and/or the distance  $h_{sub.1}$  from the first end **621** to the flange **640**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**. In the same or alternative aspects, the height of the support engaging portion **630** and/or the distance  $h_{sub.2}$  from the second end **623** to the flange **640**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**.

(59) In certain aspects, the baluster engaging portion **620** can include a sidewall **622** that extends from the flange **640** to the first end **621**, as best seen in FIG. 6A. In various aspects, the baluster engaging portion **620** has a width  $w_{sub.1}$  that can include any or all of the values and ranges of the width  $w_{sub.1}$  of the baluster engaging portion **220** of the baluster connector **210** discussed above. In various aspects, the support engaging portion **630** can have a width  $w_{sub.2}$  that includes any of the values and/or ranges of the support engaging portion **230** of the baluster connector **210** of FIG. 2A discussed above.

(60) In aspects, the support engaging portion **630** can include one or more structural features that can contact at least a portion of a bottom rail, e.g., the bottom rail **120** of FIG. 1 to temporarily or permanently secure the baluster connector **600** to a bottom rail. For instance, as best seen in the aspect depicted in FIG. 6A, the support engaging portion **630** exhibits a variable width shape having a maximum width portion **633** positioned about midway along the length of the support engaging portion **630**. In aspects, this maximum width portion **633** can interact with a portion of a bottom rail, a complementary locking member associated with a bottom rail, or both, in a manner similar to that described above in FIG. 4D with respect to the baluster connector **400**. In aspects,

for example, the maximum width portion **633** may interact with the bottom rail **462**, e.g., via the inner surface **461**, in such a manner so as to physically prevent disengagement or decoupling of the baluster connector **600** (or baluster coupled thereto) from the bottom rail **462** when an upward force is applied to the baluster connector **600** or baluster coupled thereto.

(61) FIGS. 7A-7C depict another baluster connector **700**. In aspects, the baluster connector **700** includes a baluster engaging portion **720**, a support engaging portion **730**, and a flange **740**. In aspects, the baluster connector **700** can operate similarly to the baluster connector **210** discussed above with reference to FIGS. 1 and 2A-2C. For instance, in aspects, the baluster engaging portion **720** can be inserted into a void of a baluster, e.g., the void at or near a second end **135** of the baluster **132** of FIG. 1, while the support engaging portion **730** can be coupled to a bottom rail, e.g., the bottom rail **120** of FIG. 1. In the same or alternative aspects, when in the assembled configuration, the flange **740** of the baluster connector **700** may contact an outer surface of the bottom rail, e.g., the outer surface **125** of the bottom rail **120** of FIG. 1.

(62) As best seen in the aspect depicted in FIG. 7A, the baluster **710** includes a flange **740** that is positioned between a first end **721** and a second end **723**. In aspects, the baluster engaging portion **720** can extend out from the flange **740** to the first end **721**. In the same or alternative aspects, the support engaging portion **730** can extend out from the flange **740** to the second end **723**.

(63) In various aspects, one or more of the dimensional properties of the baluster connector **210** of FIGS. 2A-2C discussed above can apply to the baluster connector **700**. For instance, the height of the baluster engaging portion **720** and/or the distance  $h_{sub.1}$  from the first end **721** to the flange **740**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**. In the same or alternative aspects, the height of the support engaging portion **730** and/or the distance  $h_{sub.2}$  from the second end **723** to the flange **740**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**.

(64) In certain aspects, the baluster engaging portion **720** can include a sidewall **722** that extends from the flange **740** to the first end **721**, as best seen in FIG. 7A. In various aspects, the baluster engaging portion **720** has a width  $w_{sub.1}$  that can include any or all of the values and ranges of the width  $w_{sub.1}$  of the baluster engaging portion **220** of the baluster connector **210** discussed above. In various aspects, the support engaging portion **730** can have a width  $w_{sub.2}$  that includes any of the values and/or ranges of the support engaging portion **230** of the baluster connector **210** of FIG. 2A discussed above.

(65) In aspects, the support engaging portion **730** can include one or more structural features that can contact at least a portion of a bottom rail, e.g., the bottom rail **120** of FIG. 1, to temporarily or permanently secure the baluster connector **700** to a bottom rail. For instance, as best seen in the aspect depicted in FIG. 7A, the support engaging portion **730** includes one or more tabs, e.g., tab **732**, that may move inward when inserting into an aperture in a bottom rail or support, and once inserted into an aperture, the tab **732** may return to its resting position and an engaging surface **733** of the tab **732** may aid in securing the support engaging portion **730** to the support or bottom rail, in aspects. For example, in such aspects, this engaging surface **733** can interact with a portion of a bottom rail, a complementary locking member associated with a bottom rail, or both, in a manner similar to that described above in FIG. 4D with respect to the baluster connector **400**. In aspects, for example, the engaging surface **733** may interact with the bottom rail **462**, e.g., via the inner surface **461**, in such a manner so as to physically prevent disengagement or decoupling of the baluster connector **700** (or baluster coupled thereto) from the bottom rail **462** when an upward force is applied to the baluster connector **700** or baluster coupled thereto. In one aspect, the baluster connector **700** can include two tabs positioned on opposing sides of the support engaging portion **730**, or in any other arrangement or position.

(66) As best seen in the aspect depicted in FIGS. 7B and 7C, the baluster connector **700** can optionally include a through-hole **750** that extends from the baluster engaging portion **720** to the

support engaging portion **730**, such that an interior portion of the baluster engaging portion **720** is in fluid communication with an interior portion of the support engaging portion **730**. In an alternate aspect, the baluster connector may not include such a through-hole thereby closing off all or part of an interior portion of the baluster engaging portion **720** to an interior portion of the support engaging portion **730**. For instance, in one alternative aspect, the flange **740** may structurally resemble a flat, solid disc shaped component, and/or may not include a void.

(67) FIGS. **8A-8C** depict another baluster connector **800**. In aspects, the baluster connector **800** includes a baluster engaging portion **820**, a support engaging portion **830**, and a flange **840**. In aspects, the baluster connector **800** can operate similarly to the baluster connector **210** discussed above with reference to FIGS. **1** and **2A-2C**. For instance, in aspects, the baluster engaging portion **820** can be inserted into a void of a baluster, e.g., the void at or near a second end **135** of the baluster **132** of FIG. **1**, while the support engaging portion **830** can be coupled to a bottom rail, e.g., the bottom rail **120** of FIG. **1**. In the same or alternative aspects, when in the assembled configuration, the flange **840** of the baluster connector **800** may contact an outer surface of the bottom rail, e.g., the outer surface **125** of the bottom rail **120** of FIG. **1**.

(68) As best seen in the aspect depicted in FIG. **8A**, the baluster **810** includes a flange **840** that is positioned between a first end **821** and a second end **823**. In aspects, the baluster engaging portion **820** can extend out from the flange **840** to the first end **821**. In the same or alternative aspects, the support engaging portion **830** can extend out from the flange **840** to the second end **823**.

(69) In various aspects, one or more of the dimensional properties of the baluster connector **210** of FIGS. **2A-2C** discussed above can apply to the baluster connector **800**. For instance, the height of the baluster engaging portion **820** and/or the distance  $h_{sub.1}$  from the first end **821** to the flange **840**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**. In the same or alternative aspects, the height of the support engaging portion **830** and/or the distance  $h_{sub.2}$  from the second end **823** to the flange **840**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**.

(70) In certain aspects, the baluster engaging portion **820** can include a sidewall **822** that extends from the flange **840** to the first end **821**, as best seen in FIG. **8A**. In various aspects, the baluster engaging portion **820** has a width  $w_{sub.1}$  that can include any or all of the values and ranges of the width  $w_{sub.1}$  of the baluster engaging portion **220** of the baluster connector **210** discussed above. In various aspects, the support engaging portion **830** can have a width  $w_{sub.2}$  that includes any of the values and/or ranges of the support engaging portion **230** of the baluster connector **210** of FIG. **2A** discussed above.

(71) In aspects, the support engaging portion **830** can include one or more structural features that can contact at least a portion of a bottom rail, e.g., the bottom rail **120** of FIG. **1**, to temporarily or permanently secure the baluster connector **800** to a bottom rail. For instance, as best seen in the aspect depicted in FIG. **8A**, the support engaging portion **830** includes one or more tabs, e.g., tabs **832a** and **832b**, that may move inward when inserting into an aperture in a bottom rail or support, and once inserted into an aperture, the tabs **832a** and **832b** may return to their resting position and respective engaging surfaces **833a** and **833b** may aid in securing the support engaging portion **830** to the support or bottom rail, in aspects. For example, in such aspects, the engaging surfaces **833a** and **833b** can interact with a portion of a bottom rail, a complementary locking member associated with a bottom rail, or both, in a manner similar to that described above in FIG. **4D** with respect to the baluster connector **400**. In aspects, for example, the engaging surfaces **833a** and **833b** may interact with the bottom rail **462**, e.g., via the inner surface **461**, in such a manner so as to physically prevent disengagement or decoupling of the baluster connector **800** (or baluster coupled thereto) from the bottom rail **462** when an upward force is applied to the baluster connector **800** or baluster coupled thereto. In one aspect, the baluster connector **800** can include one or more tabs positioned on opposing sides of the support engaging portion **830**, or in any other arrangement or

position.

(72) As best seen in the aspect depicted in FIGS. **8B** and **8C**, the baluster connector **800** can optionally include a through-hole **850** that extends from the baluster engaging portion **820** to the support engaging portion **830**, such that an interior portion of the baluster engaging portion **820** is in fluid communication with an interior portion of the support engaging portion **830**. In an alternate aspect, the baluster connector may not include such a through-hole thereby closing off all or part of an interior portion of the baluster engaging portion **820** to an interior portion of the support engaging portion **830**. For instance, in one alternative aspect, the flange **840** may structurally resemble a flat, solid disc shaped component, and/or may not include a void.

(73) FIGS. **9A-9C** depict another baluster connector **900**. In aspects, the baluster connector **900** includes a baluster engaging portion **920**, a support engaging portion **930**, and a flange **940**. In aspects, the baluster connector **900** can operate similarly to the baluster connector **210** discussed above with reference to FIGS. **1** and **2A-2C**. For instance, in aspects, the baluster engaging portion **920** can be inserted into a void of a baluster, e.g., the void in the second end **135** of the baluster **132** of FIG. **1**, while the support engaging portion **930** can be coupled to a bottom rail, e.g., the bottom rail **120** of FIG. **1**. In the same or alternative aspects, when in the assembled configuration, the flange **940** of the baluster connector **900** may contact an outer surface of the bottom rail, e.g., the outer surface **125** of the bottom rail **120** of FIG. **1**.

(74) As best seen in the aspect depicted in FIG. **9A**, the baluster **910** includes a flange **940** that is positioned between a first end **921** and a second end **923**. In aspects, the baluster engaging portion **920** can extend out from the flange **940** to the first end **921**. In the same or alternative aspects, the support engaging portion **930** can extend out from the flange **940** to the second end **923**.

(75) In various aspects, one or more of the dimensional properties of the baluster connector **210** of FIGS. **2A-2C** discussed above can apply to the baluster connector **900**. For instance, the length of the baluster engaging portion **920** and/or the distance  $h_{sub.1}$  from the first end **921** to the flange **940**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**. In the same or alternative aspects, the length of the support engaging portion **930** and/or the distance  $h_{sub.2}$  from the second end **923** to the flange **940**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**.

(76) In certain aspects, the baluster engaging portion **920** can include a sidewall **922** that extends from the flange **940** to the first end **921**, as best seen in FIG. **9A**. In various aspects, the baluster engaging portion **920** has a width  $w_{sub.1}$  that can include any or all of the values and ranges of the width  $w_{sub.1}$  of the baluster engaging portion **220** of the baluster connector **210** discussed above. In various aspects, the support engaging portion **930** can have a width  $w_{sub.2}$  that includes any of the values and/or ranges of the support engaging portion **230** of the baluster connector **210** of FIG. **2A** discussed above.

(77) In aspects, the support engaging portion **930** can include one or more structural features that can contact at least a portion of a bottom rail, e.g., the bottom rail **120** of FIG. **1**, to temporarily or permanently secure the baluster connector **900** to a bottom rail. For instance, as best seen in FIG. **9A**, the support engaging portion **930** includes a plurality of spaced apart locking members **932** that can interact with a portion of a bottom rail, a complementary locking member associated with a bottom rail, or both. For instance, in an aspect of a railing system **460** depicted in FIG. **4D**, the support engaging portion **930** can be inserted into an aperture of a bottom rail, e.g., the aperture **463** of the bottom rail **462**, and a portion of one or more of the plurality of spaced apart locking members **932**, e.g., the portion **933** depicted in FIG. **9A**, can interact with an inner surface **461** of the bottom rail **462** (and/or a complementary locking member associated with the bottom rail **462** not depicted in FIG. **4D**) in order to secure the baluster connector **900** in place and to the bottom rail **462**. In aspects, for example, the portion **933** of one of the plurality of spaced apart locking members **932** may interact with the bottom rail **462**, e.g., via the inner surface **461**, in such a

manner so as to physically prevent disengagement or decoupling of the baluster connector **900** (or baluster coupled thereto) from the bottom rail **462** when an upward force is applied to the baluster connector **900** or baluster coupled thereto.

(78) FIGS. **10A-10C** depict another baluster connector **1000**. In aspects, the baluster connector **1000** includes a baluster engaging portion **1020**, a support engaging portion **1030**, and a flange **1040**. In aspects, the baluster connector **1000** can operate similarly to the baluster connector **210** discussed above with reference to FIGS. **1** and **2A-2C**. For instance, in aspects, the baluster engaging portion **1020** can be inserted into a void of a baluster, e.g., the void at or near the second end **135** of the baluster **132** of FIG. **1**, while the support engaging portion **1030** can be coupled to a bottom rail, e.g., the bottom rail **120** of FIG. **1**. In the same or alternative aspects, when in the assembled configuration, the flange **1040** of the baluster connector **1000** may contact an outer surface of the bottom rail, e.g., the outer surface **125** of the bottom rail **120** of FIG. **1**.

(79) As best seen in the aspect depicted in FIG. **10A**, the baluster **1010** includes a flange **1040** that is positioned between a first end **1021** and a second end **1023**. In aspects, the baluster engaging portion **1020** can extend out from the flange **1040** to the first end **1021**. In the same or alternative aspects, the support engaging portion **1030** can extend out from the flange **1040** to the second end **1023**.

(80) In various aspects, one or more of the dimensional properties of the baluster connector **210** of FIGS. **2A-2C** discussed above can apply to the baluster connector **1000**. For instance, the height of the baluster engaging portion **1020** and/or the distance  $h_{sub.1}$  from the first end **1021** to the flange **1040**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**. In the same or alternative aspects, the height of the support engaging portion **1030** and/or the distance  $h_{sub.2}$  from the second end **1023** to the flange **1040**, can include any or all of the values and ranges discussed above with reference to the respective portions of the baluster connector **210**.

(81) In certain aspects, the baluster engaging portion **1020** can include a sidewall **1022** that extends from the flange **1040** to the first end **1021**, as best seen in FIG. **10A**. In various aspects, the baluster engaging portion **1020** has a width  $w_{sub.1}$  that can include any or all of the values and ranges of the width  $w_{sub.1}$  of the baluster engaging portion **220** of the baluster connector **210** discussed above. In various aspects, the support engaging portion **1030** can have a width  $w_{sub.2}$  that includes any of the values and/or ranges of the support engaging portion **230** of the baluster connector **210** of FIG. **2A** discussed above.

(82) In aspects, the support engaging portion **1030** can include one or more structural features that can contact at least a portion of a bottom rail, e.g., the bottom rail **120** of FIG. **1**, to temporarily or permanently secure the baluster connector **1000** to a bottom rail. For instance, as best seen in the aspect depicted in FIG. **10A**, the support engaging portion **1030** exhibits a variable width shape having a maximum width portion **1033** positioned about midway along the length of the support engaging portion **1030**. In aspects, this maximum width portion **1033** can interact with a portion of a bottom rail, a complementary locking member associated with a bottom rail, or both, in a manner similar to that described above in FIG. **4D** with respect to the baluster connector **400**. In aspects, for example, the maximum width portion **1033** may interact with the bottom rail **462**, e.g., via the inner surface **461**, in such a manner so as to physically prevent disengagement or decoupling of the baluster connector **1000** (or baluster coupled thereto) from the bottom rail **462** when an upward force is applied to the baluster connector **1000** or baluster coupled thereto.

(83) As discussed above, in certain aspects, the baluster connectors disclosed herein may be utilized to secure a baluster to a top rail, and/or to a top rail and a bottom rail. For instance, in the aspect depicted in FIG. **11**, a railing system **1100** includes the utilization of baluster connectors to couple balusters to a top rail **1140** and to a bottom rail **1120**. The top rail **1140** and the bottom rail **1120** can have any or all of the parameters discussed above with reference to the respective top rail **110** and bottom rail **120** of FIG. **1**. In the aspect depicted in FIG. **11**, the baluster connectors **400** and **400a**

are utilized to connect the baluster **1150** to the bottom rail **1120** and to the top rail **1140**, respectively. In aspects, the baluster connectors **400** and **400a** can have any or all of the properties of the baluster connector **400** discussed above with reference to FIGS. **4A-4C**. In alternative aspects not depicted in FIG. **11**, other baluster connectors can be utilized, e.g., such as any of the other baluster connectors disclosed herein.

(84) In the aspect depicted in FIG. **11**, the support engaging portion **430** of the baluster connector **400** can be inserted into an aperture of a bottom rail, e.g., the aperture **1110** of the bottom rail **1120**, and extend into a void **1121** of the bottom rail **1120**. In such aspects, a portion of the support engaging portion **430**, e.g., the portion **433** of one or more of the plurality of spaced apart locking members **432** depicted in FIG. **4A**, can interact with an inner surface **1122** of the bottom rail **1120** (and/or a complementary locking member associated with the bottom rail **1120** not depicted in FIG. **11**) in order to secure the baluster connector **400** in place and to the bottom rail **1120**. In aspects, for example, the portion **433** of the support engaging portion **430** may interact with the bottom rail **1120**, e.g., via the inner surface **1122**, in such a manner so as to physically prevent disengagement or decoupling of the baluster connector **400** (or baluster **1150** coupled thereto) from the bottom rail **1120** when an upward force is applied to the baluster connector **400** or baluster **1150** coupled thereto.

(85) In the aspect depicted in FIG. **11**, the support engaging portion **430a** of the baluster connector **400a** can be inserted into an aperture of a top rail, e.g., the aperture **1130** of the top rail **1140**, and extend into a void **1141** of the top rail **1140**. In such aspects, a portion of the support engaging portion **430a**, e.g., the portion **433** of one or more of the plurality of spaced apart locking members **432** of the baluster connector **400** depicted in FIG. **4A**, can interact with an inner surface **1142** of the top rail **1140** (and/or a complementary locking member associated with the top rail **1140** not depicted in FIG. **11**) in order to secure the baluster connector **400a** in place and to the top rail **1140**. In aspects, for example, the portion **433** of the support engaging portion **430a** may interact with the top rail **1140**, e.g., via the inner surface **1142**, in such a manner so as to physically prevent disengagement or decoupling of the baluster connector **400a** (or baluster **1150** coupled thereto) from the top rail **1140** when a downward force is applied to the baluster connector **400a** or baluster **1150** coupled thereto.

(86) In various aspects, a method for installing a railing system is provided. In one aspect, the method can include inserting a first end of a baluster into an aperture in a top rail. In the same or alternative aspects, the method can include inserting a support engaging portion of a baluster connector into an aperture of a bottom rail, and positioning a second end of a baluster over a baluster engaging portion of the baluster connector. In aspects, positioning the second end of the baluster over the baluster engaging portion can include sliding the baluster upward away from the bottom rail and into a void within the top rail and then back down over the baluster engaging portion of the baluster connector. In various aspects, the method can also optionally include installing a top and/or bottom rail prior to installing the baluster in the top rail; and/or the baluster connector and/or baluster in the bottom rail. In aspects, the railing system and/or components mentioned with respect to the methods of installation described herein can include any or all of the parameters of such components and systems mentioned above with respect to FIGS. **1-10C**.

(87) In various aspects, another method for installing a railing system is provided. In aspects, the method can include inserting a baluster engaging portion of a baluster connector into each of a first end and a second end of a baluster. In the same or alternative aspects, the method can include inserting a support engaging portion of each baluster connector into an aperture or void in the top and bottom rails forming a railing segment assembly. In aspects, the method can also include securing the railing segment assembly to one or more posts or supports. In aspects, the railing system and/or components mentioned with respect to the methods of installation described herein can include any or all of the parameters of such components and systems mentioned above with respect to FIGS. **1-11**.



(88) The present disclosure can be described in accordance with the following numbered clauses.

(89) Clause 1. A baluster connector, comprising: a first end; a second end opposite the first end; a flange positioned between the first end and the second end; a baluster engaging portion, wherein the baluster engaging portion extends out from the flange to the first end; and a support engaging portion, wherein the support engaging portion extends out from the flange to the second end.

(90) Clause 2. The baluster connector according to clause 1, wherein the second end is adapted to be inserted into a support.

(91) Clause 3. The baluster connector according to clause 2, wherein the support comprises a bottom rail.

(92) Clause 4. The baluster connector according to clause 2, wherein the second end, the support engaging portion, or both, comprise one or more locking members for engaging a portion of the support, a complementary locking member associated with the support, or both.

(93) Clause 5. The baluster connector according to clause 2, wherein an outer surface of the second end, an outer surface of the support engaging portion, or both, comprise one or more ridges adapted to contact at least a portion of the support to provide a friction fit between the support and at least a portion of the baluster connector.

(94) Clause 6. The baluster connector according to any of clauses 1-5, wherein the second end is positioned about 2 millimeters (mm) to about 50 mm, about 3 mm to about 45 mm, or about 4 mm to about 40 mm, from the flange.

(95) Clause 7. The baluster connector according to any of clause 1-6, wherein the first end is positioned about 2 millimeters (mm) to about 50 mm, about 3 mm to about 45 mm, or about 4 mm to about 40 mm, from the flange.

(96) Clause 8. The baluster connector according to any of clauses 1-7, wherein the first end is sized to fit in an interior portion of a baluster.

(97) Clause 9. The baluster connector according to any of clauses 1-8, wherein the baluster engaging portion has a width  $w_1$  that is from about 2.5 millimeters (mm) to about 110 mm, from about 5 mm to about 80 mm, from about 6 mm to about 65 mm, or from about 7 mm to about 50 mm.

(98) Clause 10. The baluster connector according to any of clauses 1-9, wherein the support engaging portion has a width  $w_2$  that is from about 2 millimeters (mm) to about 45 mm, from about 3 mm to about 40 mm, or from about 4 mm to about 35 mm.

(99) Clause 11. The baluster connector according to any of clauses 1-10, wherein the baluster engaging portion comprises a sidewall extending from the flange to the first end.

(100) Clause 12. The baluster connector according to any of clauses 1-11, wherein the baluster engaging portion exhibits a substantially similar width  $w_1$  from a position adjacent or at the first end to a position adjacent the flange.

(101) Clause 13. A railing system, comprising: a top rail, the top rail extending from a first end to a second end; a bottom rail, the bottom rail extending from a first end to a second end, wherein the bottom rail is vertically spaced apart from the top rail; one or more balusters, each of the one or more balusters comprising a first end and a second end; and one or more baluster connectors of any of clauses 1 and 6-12, each of the one or more baluster connectors adapted for coupling to: i) the second end of one of the one or more balusters; and ii) the bottom rail.

(102) Clause 14. The railing system according to clause 13, wherein the first end of each of the one or more balusters is adapted for insertion into at least a portion of the top rail, and wherein the baluster engaging portion of each of the one or more baluster connectors extends out from the flange to the baluster connector first end a distance that is equal to or greater than a distance the baluster first end can travel within a void of the top rail, when the railing system is in an assembled configuration.

(103) Clause 15. The railing system according to clause 13 or 14, wherein, when the railing system is in an assembled configuration, the support engaging portion of each of the one or more baluster

connectors comprises one or more locking members that engage at least a portion of the bottom rail, a complementary locking member associated with the bottom rail, or both.

(104) Clause 16. The railing system according to any of clauses 13-15, wherein, when the railing system is in an assembled configuration, an outer surface of the support engaging portion of each of the one or more baluster connectors comprises one or more ridges adapted to contact at least a portion of the bottom rail to provide a friction fit within the at least a portion of the bottom rail.

(105) Clause 17. The railing system according to any of clauses 13-16, wherein an outer surface of the flange contacts an outer surface of the bottom rail, when the railing system is in an assembled configuration.

(106) Clause 18. The railing system according to any of clauses 13-17, wherein the baluster engaging portion of each of the one or more baluster connectors is positioned at least partly in an interior portion of a baluster of the one or more balusters, when the railing system is in an assembled configuration.

(107) Clause 19. The railing system according to any of clauses 13-18, wherein the support engaging portion of each of the one or more baluster connectors is positioned in a void in the bottom rail, when the railing system is in an assembled configuration.

(108) Clause 20. The railing system according to any of clauses 13-19, wherein the railing system is adapted for installation on a level surface, an angled surface, or a combination thereof.

(109) From the foregoing, it will be seen that this invention is one well adapted to attain all the ends and objects hereinabove set forth together with other advantages which are obvious and which are inherent to the structure.

(110) It will be understood that certain features and subcombinations are of utility and may be employed without reference to other features and subcombinations. This is contemplated by and is within the scope of the claims.

(111) While specific elements and steps are discussed in connection to one another, it is understood that any element and/or steps provided herein is contemplated as being combinable with any other elements and/or steps regardless of explicit provision of the same while still being within the scope provided herein. Since many possible embodiments may be made of the disclosure without departing from the scope thereof, it is to be understood that all matter herein set forth or shown in the accompanying drawings is to be interpreted as illustrative and not in a limiting sense.

## Claims

1. A railing system, comprising: a top rail, the top rail extending from a first end to a second end; a bottom rail, the bottom rail extending from a first end to a second end, wherein the bottom rail is vertically spaced apart from the top rail, and wherein the bottom rail includes a first surface vertically spaced apart from a second surface, the first surface having an outer surface, the outer surface of the first surface facing the top rail, and an opposing inner surface, the inner surface of the first surface facing the second surface; one or more balusters, each of the one or more balusters comprising a first end and a second end; one or more baluster connectors comprising: a first end; a second end opposite the first end; a flange positioned between the first end and the second end; a baluster engaging portion, wherein the baluster engaging portion extends out from the flange to the first end; a support engaging portion, wherein the support engaging portion extends out from the flange to the second end, an outer surface of the support engaging portion comprising one or more circular ridges; and when in an assembled configuration the one or more circular ridges of the outer surface of the support engaging portion are in contact with at least a portion of the first surface of the bottom rail.

2. The railing system according to claim 1, wherein the first end of each of the one or more balusters is adapted for insertion into at least a portion of the top rail, and wherein, when the railing system is in the assembled configuration, the baluster engaging portion of each of the one or more

- baluster connectors extends out from the flange to the first end of the one or more baluster connectors a first distance that is equal to or greater than a second distance the first end of the one or more balusters can vertically shift within a void of the top rail.
3. The railing system according to claim 1, wherein, when the railing system is in the assembled configuration, the support engaging portion of each of the one or more baluster connectors comprises one or more locking members in contact with the inner surface of the first surface of the bottom rail.
  4. The railing system according to claim 1, wherein an outer surface of the flange is in contact with the outer surface of the first surface of the bottom rail, when the railing system is in the assembled configuration.
  5. The railing system according to claim 4, wherein the baluster engaging portion of each of the one or more baluster connectors is positioned at least partly in an interior portion of a baluster of the one or more balusters, when the railing system is in an assembled configuration.
  6. The railing system according to claim 5, wherein the support engaging portion of each of the one or more baluster connectors is positioned between the first surface of the bottom rail and the second surface of the bottom rail, when the railing system is in the assembled configuration.
  7. The railing system according to claim 1, wherein the railing system is adapted for installation on a level surface, an angled surface, or a combination thereof.
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